

FemPlate: finite-element modal synthesis of plates and membranes of arbitrary shape

Technical documentation

Olivier Doaré*

July 2026 — FemPlate v0.1

Abstract

FemPlate is an audio plugin that simulates the linear and weakly nonlinear vibration of a thin plate (or tensioned membrane) of arbitrary user-drawn shape, with mixed boundary conditions. The plate eigenmodes are computed by a non-conforming (Morley) finite-element discretisation and a dense subspace eigensolver, both implemented in the reusable, JUCE-free library `FxmeTools/acoustics`. Sound is produced by a bank of resonant band-pass filters, one per mode, following the modal-synthesis approach of the MechanOdd project. A first-order tension law allows audio-rate retuning of the whole bank without re-solving the eigenproblem, and this same mechanism supports an efficient approximation of geometric (von Kármán–Berger) nonlinearity: amplitude-dependent pitch glide and a cubic inter-mode energy-cascade surrogate, both realtime-safe and unconditionally stable. This document details the mathematical model, the discretisation, the eigensolution, the audio implementation and the nonlinear approximations, with references to the literature.

Contents

1	Continuous model	1
1.1	Scaled Kirchhoff plate with tension	1
1.2	Boundary conditions	2
2	Finite-element discretisation	2
2.1	Mesh generation	2
2.2	The Morley element	2
2.3	Element matrices	3
2.4	Validation	3
3	The eigenvalue problem	3
3.1	Formulation and solver	3
3.2	Audio-rate tension changes without re-solving	5
4	Modal synthesis with resonant filter banks	6
4.1	Modal decomposition	6
4.2	Frequency and damping laws	6
4.3	Excitation, pickup, gain compensation	7

*FX-Mechanics, github.com/odoare. Model, implementation and documentation developed with Claude (Anthropic).

5	Nonlinear approximation	7
5.1	Dynamic tension: the Berger approximation	7
5.2	Mode cascade: a multi-band cubic ladder	8
5.3	The statistical mode tail	10
5.4	Relation to other nonlinear plate synthesis	11
6	Implementation summary	12

1 Continuous model

1.1 Scaled Kirchhoff plate with tension

The transverse deflection $w(\mathbf{x}, t)$ of a thin plate occupying a domain $\Omega \subset \mathbb{R}^2$ is modelled by Kirchhoff–Love theory [1, 2] augmented by a uniform in-plane tension N and two damping mechanisms:

$$\rho h \ddot{w} + c_1 \dot{w} + c_2 \Delta^2 \dot{w} + D \Delta^2 w - N \Delta w = f(\mathbf{x}, t), \quad D = \frac{Eh^3}{12(1 - \nu^2)}, \quad (1)$$

where ρh is the surface density, D the flexural rigidity, ν the Poisson ratio ($\nu = 0.3$ in FemPlate), c_1 a viscous (air/support) damping coefficient and c_2 a stiffness-proportional (material, Kelvin–Voigt-type) damping coefficient. Dividing by ρh and rescaling space and time so that the coefficients of $\ddot{w}$ and $\Delta^2 w$ are unity yields the *scaled* equation actually used throughout the code:

$$\ddot{w} + c_v \dot{w} + c_m \Delta^2 \dot{w} + \Delta^2 w - T \Delta w = f, \quad (2)$$

with a single dimensionless *tension-to-flexural-stiffness* parameter $T \geq 0$. $T = 0$ is a pure plate; $T \rightarrow \infty$ approaches the membrane limit. Frequencies obtained from (2) are dimensionless; the plugin maps the first eigenfrequency onto the user parameter f_1 and all others follow proportionally (§4), so the absolute material constants never need to be specified.

1.2 Boundary conditions

The boundary $\partial\Omega$ is split by the user into segments, each carrying one of four classical conditions:

Name	Essential constraints	Natural conditions	DOFs constrained
Free	—	$M_n = 0, V_n = 0$	—
Simple support	$w = 0$	$M_n = 0$	vertex w
Clamp	$w = 0, \partial_n w = 0$	—	vertex w , edge $\partial_n w$
Sliding support	$\partial_n w = 0$	$V_n = 0$	edge $\partial_n w$

where M_n and V_n are the bending moment and effective (Kirchhoff) shear. Only the *essential* conditions are enforced; the natural ones are satisfied weakly by the variational formulation [3]. The last column anticipates §2.2: the four conditions map one-to-one onto the degrees of freedom of the Morley element, which is the main reason this element was chosen.

2 Finite-element discretisation

2.1 Mesh generation

The library meshes an arbitrary closed polygon (the user’s outline, sampled from a closed Catmull–Rom spline [4] for freehand shapes) with a target element size h :

1. *Boundary resampling.* The outline is walked at arc-length spacing $\approx h$. Vertices where the outline turns by more than 30° are detected as corners and kept exactly, so rectangles remain rectangles. Every boundary sample stores its arc-length parameter $t \in [0, 1)$, which is later used to attach the per-segment boundary conditions.
2. *Interior lattice.* A hexagonal lattice of pitch h fills the bounding box; points closer than $0.62h$ to the boundary or outside the polygon are discarded. A small deterministic jitter ($\pm 0.06h$) breaks the cocircular degeneracies that regular lattices would otherwise present to the triangulator.
3. *Triangulation.* Bowyer–Watson incremental Delaunay insertion [5, 6], with a super-triangle and cavity re-triangulation. The implementation is the plain $O(n^2)$ variant, entirely adequate for the $n \lesssim 2000$ points used here.
4. *Clipping.* Triangles whose centroid falls outside the polygon are removed (this handles non-convex outlines), orientation is normalised to counter-clockwise, and an edge table with adjacency is built. Edges with a single adjacent triangle are boundary edges; they inherit the wrap-aware midpoint of their endpoints’ arc parameters.

2.2 The Morley element

The biharmonic operator requires C^1 continuity for conforming elements, which leads to high-order elements (Argyris, Bell) that are expensive and delicate. FemPlate instead uses the *Morley triangle* [7], the simplest non-conforming plate bending element: on each triangle the deflection is a full quadratic polynomial (6 coefficients) determined by 6 degrees of freedom,

- the deflections w_i at the three vertices,
- the normal derivatives $(\partial_n w)_e$ at the three edge midpoints.

The element is non-conforming (w is discontinuous along edges except at vertices), yet it converges for fourth-order problems, and its eigenvalue approximations converge at rate $O(h^2)$, *from below* [8, 9] — both properties are verified numerically in §2.4. Crucially for FemPlate, its two DOF families are exactly the quantities constrained by the four boundary conditions of §1.2.

Rather than hard-coding the classical closed-form shape functions, each element inverts a 6×6 generalized Vandermonde system: writing the local polynomial in centroid-centred monomials $p(\mathbf{x}) = \sum_m c_m \mu_m(\mathbf{x})$, $\{\mu_m\} = \{1, X, Y, X^2, XY, Y^2\}$, the matrix $C_{im} = \ell_i(\mu_m)$ of the six DOF functionals ℓ_i is inverted by Gaussian elimination with partial pivoting; column j of C^{-1} holds the monomial coefficients of shape function N_j . Centroid-centred coordinates keep C well-conditioned at the element scale. The mid-edge normal is the *global* normal of the edge (stored with $v_0 < v_1$, normal to the left of $v_0 \rightarrow v_1$), so the shared DOF has the same meaning in both adjacent elements.

2.3 Element matrices

With constant curvatures per element (second derivatives of a quadratic), the bending stiffness is exact:

$$K_{ij}^e = A_e \boldsymbol{\kappa}_i^\top \mathbf{D} \boldsymbol{\kappa}_j, \quad \boldsymbol{\kappa}_j = \begin{pmatrix} \partial_{xx} N_j \\ \partial_{yy} N_j \\ 2 \partial_{xy} N_j \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix}, \quad (3)$$

(A_e = element area; flexural rigidity is 1 after scaling). The tension (geometric) matrix and the consistent mass matrix are integrated with quadrature rules exact for their integrands:

$$G_{ij}^e = \int_e \nabla N_i \cdot \nabla N_j \, dA \quad \text{3-point midpoint rule (degree 2, exact),} \quad (4)$$

$$M_{ij}^e = \int_e N_i N_j \, dA \quad \text{6-point degree-4 rule [11] (quartic, exact).} \quad (5)$$

G is the discrete counterpart of $-\Delta$ restricted to the plate space and doubles as the *geometric stiffness* familiar from buckling analysis [3]. Global assembly scatters K^e, G^e, M^e into dense symmetric matrices over the *free* DOFs only: essential boundary conditions (§1.2) are applied by elimination, using each boundary vertex's / edge's arc parameter to look up its segment's condition. At a junction between two segments the vertex takes the strongest condition (clamp > simple support > sliding > free).

2.4 Validation

The console test suite (`Tests/FemTests.cpp`) verifies the implementation against classical analytical results [1]:

- simply supported unit square: $\lambda_{mn} = \pi^4(m^2 + n^2)^2$, with the measured $O(h^2)$ convergence from below (error ratio 3.9 when h is halved) and the exact tension coefficient $g_{11} = 2\pi^2$ (§3.2);
- clamped circle: frequency ratios 2.081, 3.414, 3.893 and the absolute fundamental $\omega_{01} = (\beta_{01}/R)^2$, $\beta_{01}^2 = 10.2158$ [1], within 2–5% at moderate mesh density;
- mixed clamped/free boundaries, mode-shape vanishing on the clamped side, and the first-order tension law against an exact re-solve.

3 The eigenvalue problem

3.1 Formulation and solver

Discretisation of (2) (undamped part) yields the symmetric generalized eigenproblem

$$(K + T_0 G) \varphi_k = \lambda_k M \varphi_k, \quad \lambda_k = \omega_k^2, \quad \varphi_k^T M \varphi_l = \delta_{kl}, \quad (6)$$

solved at a *reference tension* T_0 (the value of the Tension knob when the user presses *Compute*). With n free DOFs ($n \approx$ vertices + edges $\approx 4 \times$ vertices, typically 300–5000 over the Grid range) and only the lowest $m \leq 256$ modes wanted (capped at $\sim n/6$ so the top modes stay resolved by the mesh), the solver is classical *subspace iteration with Rayleigh–Ritz projection* [12, 3]:

1. Form $A = K + T_0 G$ and the shifted operator $P = A + \sigma M$ with $\sigma = 10^{-5} \text{tr}(A)/\text{tr}(M)$. The shift keeps P positive definite when free (Neumann-type) boundaries leave rigid-body modes in A 's null space. P is Cholesky-factored once ($O(n^3/3)$).
2. Iterate on a block of $p = m + \max(8, m/2)$ vectors: $Z \leftarrow P^{-1} M X$ (one inverse-power step towards the low spectrum), followed by Rayleigh–Ritz on $\text{span}(Z)$: project $A_p = Z^T A Z$, $M_p = Z^T M Z$, solve the $p \times p$ generalized problem by Cholesky reduction and cyclic Jacobi [13], and take the Ritz vectors as the next block X .
3. Stop when the wanted eigenvalues are stationary: relative 10^{-6} on the lower half, 10^{-4} on the upper half — the slowest to converge, and the modes whose FEM discretisation error is orders of magnitude larger anyway (both thresholds are far below one cent in frequency); or after 60 iterations.

Implementation notes that matter for speed: the iteration block is stored *transposed* (every iteration vector contiguous in memory, so all $O(n^2p)$ loops stream cache lines), the independent per-vector solves and matrix-vector products are spread over a few worker threads, and the Jacobi solver of the projected $p \times p$ problem uses a *relative* off-diagonal convergence test (the entries scale with $\lambda^2 \sim 10^{12}$; an absolute threshold silently burns the full sweep budget on every call). Together these give a $\sim 4\times$ speed-up over the straightforward implementation (23.6 s $\rightarrow$ 5.6 s for 128 modes on a 931-DOF mesh).

Sparse storage. A Morley element couples a degree of freedom only to the five others on its own triangle, so an assembled row of K , G or M holds about eleven non-zeros — and, this being the useful part, about eleven whatever the mesh density. Dense storage of the same matrix is n per row, so the fraction it wastes grows without bound: at $n = 5000$ the three matrices are 24 million doubles standing in for 55 thousand real numbers. They are therefore assembled in compressed sparse rows, in two passes over the triangles: the first gathers nothing but the six DOF indices of each element and builds the sparsity pattern (one pattern, shared by all three matrices — same mesh, same couplings), the second is the expensive one and scatters element values into it. Both passes read the DOF indices through the same helper, so the pattern cannot fail to cover what the assembly scatters.

The shifted operator $P = A + \sigma M$ is still factorised *densely*, which leaves one n^2 allocation and the $O(n^3/3)$ factorisation in place. The working set is thus

$$\underbrace{8n^2}_{\text{Cholesky factor}} + \underbrace{32pn}_{\text{iteration block}} + \underbrace{O(n)}_{\text{sparse operators}} \text{ bytes,} \quad \text{against } 32n^2 \text{ dense,} \quad (7)$$

a factor of three to four in practice, with the matrix-vector products of the iteration — three per vector per sweep — dropping from $O(n^2)$ to $O(11n)$. Measured on the same machine, same meshes, both paths run back to back:

n	modes	dense		sparse		gain
		time	memory	time	memory	
2149	64	13.1 s	125 MB	7.3 s	36 MB	1.8 $\times$ / 3.4 $\times$
3841	128	114.8 s	419 MB	59.9 s	122 MB	1.9 $\times$ / 3.4 $\times$
6029	128	335.5 s	1040 MB	171.0 s	287 MB	2.0 $\times$ / 3.6 $\times$

The two agree bit for bit, not merely to round-off: compressed rows keep their column indices sorted, so a sparse row walk sums the non-zeros in exactly the order a dense row walk does. The console test suite asserts that equality, which makes it sharp enough to catch an indexing slip.

What still sets the ceiling. The remaining n^2 is the dense factorisation, and it is what caps the Grid knob. Over the Grid range the projected footprint is now 5 MB at Grid 16 ($n = 554$), 24 MB at 24 ($n = 1243$), 63 MB at 32 ($n = 2195$), 132 MB at 40 ($n = 3440$) and 243 MB at the top, Grid 48 ($n = 4913$) — one third of what the same range cost dense. The editor shows the figure next to the mesh size once it passes 150 MB. Removing the ceiling altogether means factorising sparsely too: a bandwidth-reducing (reverse Cuthill–McKee [10]) renumbering brings the profile of these matrices down by more than an order of magnitude, and a profile Cholesky over that envelope needs no symbolic analysis phase. Measured on the actual DOF graph, that would take the Grid-48 factorisation from 39.5 to 0.03 Gflop and its storage from 737 to 5 MB. It is not, however, where the time goes: solving for 8 modes instead of 128 on the same mesh only drops the total from 85 to 18 s, so the factorisation is a fifth of the cost and the subspace iteration is the rest. Sparsity would make the mesh nearly free and

leave the $O(np^2)$ Rayleigh–Ritz projections — which scale with the number of modes asked for, not with the mesh — as the thing to beat.

Code layout. None of the above is about plates, and it is not written as though it were. The matrix storage, the Cholesky and Jacobi kernels and the subspace iteration live in `FxmeTools/math`; `FxmeTools/acoustics` keeps the boundary conditions, the Morley element and the modal quantities. The eigensolver never sees a matrix at all — it multiplies through a `SymmetricOperator` and solves through an `SpdSolver` — which is what makes the storage a choice at the call site (`ModalOptions::storage`, sparse by default, dense kept as the reference the sparse path is validated against) rather than a rewrite, and what will let a profile factorisation drop in as one more `SpdSolver` with nothing else moving.

Finally, the computed modal data (eigenvalues, tension coefficients, shapes, statistical tail) is cached inside the plugin state: loading a session or preset republishes the model immediately, with no recomputation. The mesh is not serialised — `generateMesh` is a pure deterministic function of the outline and density, so it is rebuilt on load and checked against the stored vertex count. Eigenvalues below an absolute threshold (10^{-4} ; rigid-body and near-rigid modes of free plates) are dropped. Eigenvectors are mass-normalised, their vertex values are exported for display and for the pointwise evaluations $\varphi_k(\mathbf{x})$ used by the synthesis (linear interpolation on the located triangle; the mid-edge DOFs remain internal). The whole computation runs on a background thread with progress reporting; completed models are published to the audio thread by an atomic pointer swap with generation-based reclamation.

3.2 Audio-rate tension changes without re-solving

The solver additionally returns, for every mode, the *tension sensitivity*

$$g_k = \varphi_k^\top G \varphi_k \left(= \int_{\Omega} |\nabla \varphi_k|^2 dA \right), \quad (8)$$

enabling the first-order frequency law used at audio rate:

$$\boxed{\omega_k^2(T) = \lambda_k + (T - T_0) g_k} \quad (9)$$

This is legitimate because T enters the operator *linearly* and the Rayleigh quotient is *stationary* at eigenvectors: perturbing T changes φ_k at first order in $(T - T_0)$, but that eigenvector error affects the eigenvalue only at *second* order. Equation (9) is the standard eigenvalue derivative $\partial \lambda_k / \partial T = \varphi_k^\top G \varphi_k$ of structural sensitivity analysis [16], a special case of classical perturbation theory [14, 15]; here it is exact at T_0 with $O((T - T_0)^2)$ error. For the simply supported square the law is exact for all T (the eigenvectors do not depend on T); in general, pressing *Compute* re-anchors T_0 when the knob has travelled far. The validation suite checks (9) against an exact re-solve at $T = 50$ (error $< 1\%$).

Beyond the knob itself, (9) is the enabling mechanism for the nonlinear glide of §5.1: *dynamic* tension is just a fast-moving T .

4 Modal synthesis with resonant filter banks

4.1 Modal decomposition

Expanding $w(\mathbf{x}, t) = \sum_k q_k(t) \varphi_k(\mathbf{x})$ in (2) and using M -orthonormality decouples the dynamics into damped oscillators

$$\ddot{q}_k + 2\zeta_k \omega_k \dot{q}_k + \omega_k^2 q_k = \varphi_k(\mathbf{x}_h) F(t), \quad w(\mathbf{x}_o, t) = \sum_k \varphi_k(\mathbf{x}_o) q_k(t), \quad (10)$$

for a point force F at $\mathbf{x}_h$ and a pickup at $\mathbf{x}_o$ — the standard modal synthesis framework [17, 18, 19, 20]. FemPlate realises each oscillator as a second-order IIR resonator, exactly as in the MechanOdd modal resonators: one constant-peak-gain band-pass biquad per mode (RBJ “cookbook” coefficients [21]) with $Q_k = 1/(2\zeta_k)$, excited by $\varphi_k(\mathbf{x}_h)F$ and weighted on output by $\varphi_k(\mathbf{x}_o)$. Up to 512 modes are active — though the bank stops at the highest one still below $0.47f_s$, which at a high f_1 is a small fraction of them; the rest would cost a resonator each and return silence. The bank is allocation-free and retunable on the audio thread.

4.2 Frequency and damping laws

The user selects the fundamental in hertz and the bank plays at

$$f_k = f_1 \frac{\omega_k(T_{\text{eff}})}{\omega_1(T_{\text{knob}})}, \quad T_{\text{eff}} = T_{\text{knob}} + T_{\text{dyn}}, \quad (11)$$

with all ω_k given by (9). The normalisation reference matters: it is the fundamental at the *static* knob tension, not the instantaneous one. The static Tension knob therefore keeps mode 1 pinned at f_1 and only reshapes the overtone ratios (the timbre morphs from plate towards membrane), whereas the *dynamic* tension of §5.1 shifts the whole spectrum, fundamental included — the audible hardening glide. Normalising by the instantaneous fundamental instead would cancel the glide of mode 1 and leave only the overtone-ratio compression towards the membrane spectrum (g_k/g_1 ratios are lower than the plate ratios $\sqrt{\lambda_k/\lambda_1}$), which is perceived as a *softening* — the opposite of the physical behaviour. So the timbre (inharmonic pattern) is set by geometry, boundary conditions and tension, while f_1 transposes it. Modes falling outside $[20\text{ Hz}, 0.47f_s]$ are muted.

Projecting the two damping terms of (2) onto mode k gives modal damping ratios (using $\Delta^2\varphi_k \approx \omega_k^2\varphi_k$ for the material term)

$$\zeta_k = \underbrace{\frac{c_v}{2\omega_k}}_{\text{viscous}} + \underbrace{\frac{c_m \omega_k}{2}}_{\text{material}} = \frac{\zeta_v}{\nu_k} + \zeta_m \nu_k, \quad (12)$$

where the knobs ζ_v, ζ_m are simply the two contributions to the damping ratio of *mode 1*. Viscous damping therefore lets high modes ring relatively longer (constant absolute bandwidth), material damping kills them faster — the two classical, perceptually distinct decay profiles of struck plates [22, 23].

4.3 Excitation, pickup, gain compensation

A mouse hit at $\mathbf{x}_h$ launches a half-sine force pulse (a standard hammer/impact model [22])

$$F(t) = F_0 \sin(\pi t/\tau), \quad 0 \leq t \leq \tau, \quad (13)$$

with duration τ (*Hammer* knob, 0.1–50 ms: short = bright, long = mellow, since the pulse spectrum rolls off above $\approx 1/\tau$) and amplitude F_0 (*Force* knob $\times$ click velocity). Up to 8 hammers overlap polyphonically, each carrying its own weight vector $\varphi_k(\mathbf{x}_h)$. The external audio input is injected identically at the last hit point (*effect mode*), and the output point $\mathbf{x}_o$ is a pair of automatable parameters.

Because a constant-peak band-pass transmits energy proportionally to its bandwidth ($\propto \zeta_k$), lightly damped modes would ring much quieter than damped ones. Each mode is therefore compensated by $c_k = \sqrt{\zeta_{\text{ref}}/\zeta_k}$ (clamped to $[0.5, 32]$, $\zeta_{\text{ref}} = 10^{-2}$), the loudness compensation inherited from the MechanOdd modal resonators:

$$c_k = \sqrt{\zeta_{\text{ref}}/\zeta_k}. \quad (14)$$

It equalises each mode’s ring energy rather than its peak, which has a consequence for anything driven from the output signal — see §5.2. Retunes triggered by parameter changes rewrite only the biquad coefficients (the states are preserved), which is inaudible for the small throttled steps used here.

5 Nonlinear approximation

Large-amplitude plate vibration is governed by the von Kármán equations [24, 25]: transverse deflection stretches the mid-plane, producing an amplitude-dependent membrane stress that (i) raises all frequencies (the gong/tom *pitch glide*) and (ii) couples the modes, cascading energy towards high frequencies (the cymbal *crash*, up to wave turbulence [39]). Exact modal von Kármán synthesis requires the cubic coupling tensors Γ_{klmn} — $O(N^3)$ – $O(N^4)$ storage and work [29, 38] — which is beyond a realtime budget at a few hundred modes. FemPlate implements two lightweight surrogates, one for each phenomenon; §5.4 compares them with the direct simulation of the full system, which is affordable in real time on a rectangle.

5.1 Dynamic tension: the Berger approximation

Berger [26] observed that for many plate problems the von Kármán membrane coupling is well approximated by a *spatially uniform* tension proportional to the integrated stretching:

$$\ddot{w} + \Delta^2 w - \underbrace{\left(T_{\text{knob}} + \kappa \int_{\Omega} |\nabla w|^2 dA\right)}_{T_{\text{dyn}}} \Delta w = f. \quad (15)$$

In modal coordinates $\int |\nabla w|^2 = \sum_{kl} q_k q_l \boldsymbol{\varphi}_k^T G \boldsymbol{\varphi}_l \approx \sum_k g_k q_k^2$: the stretching integral is measured by the very same G and g_k of §3.2, and the resulting uniform T_{dyn} feeds straight into the retune law (9). The Berger model captures the hardening (glide-up-and-relax) behaviour of plates and membranes and is a standard efficient approximation in the nonlinear vibration literature [26, 27]; its perceptual signature on percussion is documented by Legge & Fletcher and Rossing [28, 23].

Amplitude estimation and calibration. The stretching integral needs no field reconstruction: the modes are mass-orthonormal, so $\int_{\Omega} w^2 dA = \sum_k q_k^2$ exactly and $\int_{\Omega} |\nabla w|^2 dA \approx \sum_k g_k q_k^2$, one multiply-add per mode and per sample. What the filter bank holds is not q_k , however. A 0 dB-peak band-pass fed the true modal force is exactly $y_k = 2\zeta_k \omega_k \dot{q}_k$ — a *velocity*, normalised — so

$$q_k = \frac{y_k}{2\zeta_k \omega_k^2}, \quad \gamma = \min\left(\gamma_{\text{max}}, \alpha \cdot \text{nonlin} \cdot \left\langle \sum_k W_k y_k^2 \right\rangle\right), \quad W_k = \frac{g_k}{g_1} \left(\frac{\zeta_{\text{ref}}}{\zeta_k}\right)^2 \frac{1}{\nu_k^4}, \quad (16)$$

with $T_{\text{dyn}} = \gamma \omega_1^2 (T_{\text{knob}})/g_1$, $\alpha = 500$, $\gamma_{\text{max}} = 4$, and $\langle \cdot \rangle$ a 20-ms one-pole envelope. The missing $1/(4\zeta^2 \omega^4)$ — ten decades across the bank — is why a naive $\sum_k g_k \langle y_k^2 \rangle$ underestimates the stretching by orders of magnitude. Being derived from the velocity, each mode is a quarter period ahead of its true displacement; the cross terms $k \neq l$ and that phase both average out in the envelope.

By construction $\omega_1^2 \mapsto \omega_1^2 (1 + \gamma)$ exactly, and every other mode follows its own physical sensitivity g_k : γ is plate-independent ($\gamma = 0.4$ always means “mode 1 up ≈ 3 semitones”). It is also *damping*-independent by construction: after an impulse $y_k \sim 2\zeta_k \omega_k$, so ζ_k cancels in $W_k y_k^2$ and the peak driver varies by less than 12% over $\zeta_v, \zeta_m \in [10^{-5}, 10^{-3}]$ on the default plate. And being a quantity integrated over Ω , it is the same wherever one listens: the glide belongs to the plate, not to the pickup.

Realtime scheduling. γ is updated every 32 samples, slewed by a factor 0.2 per update ($\approx$ ms-scale smoothing), and the biquads are rewritten only when mode 1 would move by more than ≈ 2 cents ($|\Delta\gamma| > 0.0024(1 + \gamma)$, since $\Delta\omega/\omega = \frac{1}{2}\Delta\gamma/(1 + \gamma)$). Coefficient rewrites preserve filter states; the throttling keeps them free of zipper artefacts, following the same strategy as MechanOdd’s gate-morphed damping.

5.2 Mode cascade: a multi-band cubic ladder

A memoryless cubic of a signal band-limited to $[0, B_\omega]$ produces inter-modulation products only up to $3B_\omega$, concentrated near the original band. A single “distort the low modes, inject into the high ones” pass therefore *cannot* reach the top of the bank: with sources below $\sim 12f_1$ the products die out by ~ 3 kHz while the bank extends to 9 kHz, and all that remains audible of such a scheme is the side effect of its overlap floor — high-frequency damping. The physical cascade climbs the spectrum *iteratively*: products beget products, stage after stage [39].

FemPlate reproduces this with a *windowed* band ladder. The active modes are split into $N_b = 8$ contiguous index bands $\mathcal{B}_0 \dots \mathcal{B}_7$ (equal index ranges are equal frequency ranges, plate modal density being constant); band b is driven by the cubic of the W bands directly below it only (default $W = 4$),

$$w^{(b)}(t) = \frac{S}{\sqrt{|\Omega|}} \sum_{c=b-W}^{b-1} \frac{1}{n_c} \sum_{k \in \mathcal{B}_c} v_k y_k(t), \quad F_b(t) = A g_b(t) \tanh^3(B w^{(b)}(t)), \quad (17)$$

fed to mode $l \in \mathcal{B}_b$ with weight $\varphi_l(\mathbf{x}_h)/c_l$ (dividing by the bandwidth compensation c_l , which is re-applied on output, makes the audible cascade level independent of the damping settings). The source is the plate’s *motion*, over the whole plate: $v_k = \zeta_{\text{ref}}/(\zeta_k \nu_k)$ is the modal velocity scale, $n_c = \sum_{k \in \mathcal{B}_c} 1/\nu_k$ normalises each band, $1/\sqrt{|\Omega|}$ is what a mass-normalised φ_k is typically worth, and $S = 3$ is the one global gain (§5.2). The source *window* matters: summing all lower bands would let the loud strike band drive the top bands quasi-directly and the whole spectrum lights up within milliseconds; with the window, each stage extends the spectral reach by at most a factor 3 and energy genuinely climbs rung by rung. The per-band gate $g_b(t) \in [0, 1]$ adds *transfer inertia*: it follows the source presence through an attack/release envelope whose attack time grows with the band’s height on the ladder ($\tau_b = 30 \text{ ms} \times b$ and a 2 s release in the voiced defaults), so the brightening is a progressive glow and the pumping tails off smoothly instead of collapsing with the cube of the decaying ring. Voicing found a hard-saturating transfer most convincing: a large drive ($B = 16$) with a small injection gain ($A = 1.1$, capped at 1.5 — beyond that the level becomes unpleasant). A cubic is the leading nonlinearity of von Kármán dynamics [29]; at low amplitude (17) vanishes as w^3 (the linear model is recovered exactly), and at high amplitude the bands brighten in sequence, reproducing the amplitude-gated crash of cymbals and gongs [23]. All cascade internals (A , B , the source window, the gate times and the overlap floor below) are exposed as plugin parameters while the effect is being voiced; the values quoted here are the defaults. Note also that the material damping law $\zeta \propto \nu$ dominates the high-mode decay when hunting long shimmer: at $\zeta_m = 10^{-4}$ a mode at $\nu = 80$ decays in milliseconds whatever the cascade does.

Modal-overlap floor. The cubic’s inter-modulation products are broadband, but at typical plate dampings the receiving resonances are needles: bandwidth $2\zeta_l f_l$ of a few hertz against a mean mode spacing $\Delta f \approx 0.64f_1$ (the plate Weyl law gives a constant modal density $dN/d\nu = \pi/2$ for the unit square), i.e. a modal overlap of order 0.1. Most of the injected energy then falls *between* modes and is rejected — heard as distortion of isolated partials rather than a cascade, whereas the crash regime of real cymbals has overlap $\gg 1$ [23]. The target bandwidths are therefore floored, proportionally to the cascade control, at an adjustable fraction (*Overlap*,

default 0.1) of the *local* mode spacing (measured on the actual f_k array),

$$\zeta_l \geq \text{cascade} \times \text{Overlap} \times \frac{\frac{1}{2}(f_{l+1} - f_{l-1})}{2f_l}, \quad l \notin \mathcal{B}_0, \quad (18)$$

turning the receiving comb into a quasi-continuum when the cascade is engaged (and, physically consistently, shortening the high modes' ring — nonlinear energy spreading is itself a loss channel for individual modes). At cascade = 0 the linear model is untouched.

Why the source is the motion and not the sound. The obvious source signal for (17) is the one already being summed for the output, $\varphi_k(\mathbf{x}_o) c_k y_k$ — and that is precisely what it must not be. Two things stand between the audible signal and the plate's motion, and the ladder's cubic would raise both to the third power.

The first is the loudness compensation (14). A band-pass ring has peak $2\zeta_k \omega_k$, so the audible peak of a struck mode is

$$c_k \cdot 2\zeta_k \omega_k = 2\omega_k \sqrt{\zeta_{\text{ref}} \zeta_k} \propto \sqrt{\zeta_k} : \quad (19)$$

c_k equalises each mode's ring *energy* by trading duration for level, so in the audible signal a more damped mode peaks *higher*. That is sound design, and it is the right choice for what reaches the ear, but it is not what the plate does: a struck plate's modal amplitude is set by the impulse, and damping can only take energy away. Weighting the source by $v_k \propto 1/(\zeta_k \nu_k)$ removes the damping entirely, since $y_k \propto \zeta_k$; what is left is the modal velocity.

The per-band divisor n_c then supplies the *frequency* balance the ladder needs, which (19) had been providing as a side effect: physical modal amplitudes fall steeply with frequency, and a raw velocity sum outweighs band 1 with band 0 by a factor ~ 900 , starving the upper rungs. n_c is fixed by the mode indices alone, so no damping re-enters.

The second is the pickup. A mode nodal at $\mathbf{x}_o$ contributes nothing to the audible signal however hard the plate is moving there, and a real plate's cubic coupling does not know where the microphone is. The source weight is therefore the constant $1/\sqrt{|\Omega|}$, the typical size of a mass-normalised mode shape ($\int_{\Omega} \varphi_k^2 dA = 1$), which holds the drive's level on any geometry while tying it to no listening position: across five pickup positions the high-mode motion the cascade creates is identical to the last digit. What the pickup changes is what one *hears* of that motion, which is what a pickup is for. The *injection* does keep $\varphi_l(\mathbf{x}_h)$: the hit point is where the plate deflects most, the one position the nonlinearity has a physical claim to, and it is what ties the cascade's character to how the plate is played.

The Cascade knob therefore acts directly, with no damping correction on it. `Tests/CascadeMeasure.cpp` renders one strike through the real synthesis and asserts what the construction promises — the drive is damping-free, so any real spread would mean a damping term had crept back into it:

Shimmer at 300 ms, spread over	dB
Viscous $\zeta_v \in [10^{-5}, 10^{-3}]$	5.0
Material $\zeta_m \in [7 \cdot 10^{-6}, 3 \cdot 10^{-4}]$	6.6
Output point, five positions (drive, measured on the modes)	0.00

The few dB left on the damping rows are not drive: the targets ring for longer or shorter with the damping, and the middle rungs are sources for the upper ones. Meanwhile the Cascade knob spans 18 dB over its travel and a force-10 hit shimmers 45 dB more than a force-1 one — the effect stays amplitude-gated, which is the part that *is* physical.

Depletion (low-frequency loss). The transfer (17) is additive: it copies energy upward but takes none from the sources, so the struck low modes ring on untouched under the shimmer — audibly two disconnected layers. In the real cascade the low modes *lose* the energy they hand

upward. FemPlate models this as an extra dissipation on the source bands: each sample, the filter states of the modes in band b are scaled by

$$d_b = 1 - \text{Deplete} \times \text{cascade} \times \varepsilon_0 \bar{g}_b^\uparrow, \quad \varepsilon_0 = 3 \cdot 10^{-4} \quad (\text{Deplete} = 0.07 \text{ by default}), \quad (20)$$

where $\bar{g}_b^\uparrow$ is the mean gate level of the bands fed by band b : the lows decay exactly when, and as strongly as, they are pumping the highs, gluing the two ends of the spectrum. The mechanism is purely dissipative (it only removes energy), so it cannot affect stability.

Stability by structure. The obvious way to spread energy upward is a feedback loop closed through the bank’s own output, and it does not survive contact with the gains involved. Around mode k ’s resonance such a loop transmits with $|\varphi_k(\mathbf{x}_h)| \cdot |\varphi_k(\mathbf{x}_o)| \cdot c_k$, and the compensation $c_k \leq 32$ must not be forgotten: an unnormalised loop is super-critical and latches into a full-scale limit cycle the moment the cubic’s slope wakes up. Bounding the loop slope by the small-gain theorem [42, 43] restores stability but is so conservative (the bound assumes worst-case coherent feedback at the single worst mode) that the effect becomes inaudible. The ladder (17) dissolves the dilemma structurally: band b receives only from bands $c < b$ and contributes only to bands $c > b$, so the transfer graph over bands is a strict DAG — no loop exists at any stage, stability is unconditional, and the injection gain A is free to be set for audibility ($A = \text{Amp} \times \text{cascade}$, the $\tanh^3$ keeping every stage bounded).

5.3 The statistical mode tail

Even a 256-mode FEM bank remains a sparse comb at the top of the spectrum, and solving for many hundreds of accurate FEM modes is prohibitive (§3) — while a convincing crash needs a dense receiving continuum up to Nyquist. FemPlate therefore fills the bank (to 512 modes in all) with *synthetic* modes above the FEM set, built from asymptotics instead of the discrete operator:

- *Frequencies*: the plate Weyl law gives a constant modal density in ω ; the tail continues at the spacing $\Delta\omega$ measured on the top half of the FEM spectrum, with gaps jittered by $\pm 40\%$ (deterministically seeded) to avoid periodicity artefacts.
- *Tension sensitivities*: for an asymptotic mode of wavenumber κ , $\lambda = \kappa^4$ and $g = \kappa^2$, hence $g = \sqrt{\lambda}$ exactly (verified by the simply supported square, where $g_{mn} = \pi^2(m^2+n^2) = \sqrt{\lambda_{mn}}$) — so the tail obeys the tension law (9) and the Berger glide like any FEM mode.
- *Shapes*: high-frequency eigenfunctions of irregular domains behave like superpositions of random plane waves (Berry’s random-wave model [40]); each tail mode carries one, $\varphi(\mathbf{x}) = \sqrt{2/A} \cos(\boldsymbol{\kappa} \cdot \mathbf{x} + \psi)$ with $|\boldsymbol{\kappa}| = \lambda^{1/4}$, random direction and phase, and the $\sqrt{2/A}$ prefactor reproducing the mass normalisation ($\overline{\varphi^2} = 1/A$). Hit and pickup positions therefore still modulate the tail response.

The tail modes enter the synthesis exactly like FEM modes (hammer excitation, damping law, cascade bands, dynamic tension); only the contour display is FEM-only, since tail modes have no mesh shape. The identity-defining low modes stay exact, and the top of the bank becomes the quasi-continuum the cascade ladder needs — the reasoning is that of statistical energy analysis [41]: above a few dozen mode indices, individual eigenfrequencies are perceptually interchangeable and only their density and statistics matter.

5.4 Relation to other nonlinear plate synthesis

The closest published work to FemPlate’s objective is the real-time gong synthesis of Bilbao, Webb, Wang and Ducceschi [30], which reaches the same sounds — pitch glide, crash, swell —

from the opposite direction, and the contrast is worth stating plainly because it delimits what the surrogates of §5.1 and §5.2 can and cannot claim.

Direct time-domain simulation of the von Kármán plate for synthesis goes back to Bilbao’s offline work [31, 38]; what [30] adds is the algorithmic work needed to bring it inside a real-time budget.

That work simulates the dynamic Föppl–von Kármán system directly: the deflection w and the Airy stress function Φ are stepped on a uniform Cartesian grid, with the nonlinearity made fully explicit by a scalar auxiliary variable (energy quadratisation), a rank-one system solved in closed form by Sherman–Morrison, and a discrete energy balance that yields a provable stability condition [32, 30]. The nonlinear effects are not modelled separately: they are consequences of the one equation, so the glides come out non-uniform across partials and the crash appears on its own as the amplitude rises. The price is paid in three places:

- *Geometry.* The domain is a rectangle with simply supported conditions on both w and Φ . This is not incidental. The run-time bottleneck is the biharmonic solve $\Delta\Delta\Phi = -L(w, w)$ needed at every sample, which consumes 72–76% of the time step, and the fast solver that makes it affordable [33] exploits the Toeplitz-block-Toeplitz structure that the rectangle and those boundary conditions produce. An arbitrary outline would cost the solver, and with it the real-time budget.
- *Cost.* Reported timings are 0.15–0.57 s of computation per second of audio on a recent laptop CPU, and 0.26–0.99 s on an older one, for grids of roughly 300 to 720 interior points. FemPlate’s full bank — 256 modes with both the Berger glide and the cascade engaged — measures 0.054 s per second of audio on an Intel i7-10810U, roughly an order of magnitude less, because nothing is solved at run time: the eigenproblem was solved once, offline (§3; 0.5 s for the default plate), and what remains is a bank of biquads whose coefficients are rewritten when the tension moves.
- *Resolution coupling.* The grid spacing is tied to the sample rate by the stability condition, so spatial resolution is not a free modelling choice. In a modal scheme the mesh density and the audio rate are independent.

What is given up in exchange is exactness, and that paper names the limitation precisely: Berger’s approximation “[is] able to replicate pitch glides, but not the energy cascade to high frequencies characteristic of crashes” [30]. That is correct, and it is why §5.2 exists as a *separate* mechanism rather than falling out of §5.1. Two consequences follow. First, the glide here is driven by a single scalar T_{dyn} : every mode follows it through its own g_k , so the pattern of glides is fixed by geometry and boundary conditions, where in a von Kármán solution mode k ’s frequency shift depends on which modes actually carry the energy and therefore evolves through the decay. Second, the cascade of §5.2 is a caricature of the coupling, not the coupling: it reproduces the phenomenon — energy climbing the spectrum rung by rung, gated by amplitude — with a band ladder whose gains are voiced by ear, and it carries no energy balance. Stability comes from the transfer graph being acyclic rather than from a numerical invariant.

The exact modal route sits between the two. Projecting von Kármán onto the linear modes gives cubic coupling tensors Γ_{klmn} , which is the basis of the gong and cymbal synthesis of Ducceschi and Touzé [29] and of the analytical work on circular plates and shallow shells by Touzé, Thomas and Chaigne [34, 35]; Nguyen and Touzé extend it to plates of variable thickness for cymbals [36]. That route keeps the modal picture — and would keep FemPlate’s arbitrary geometry, since the tensors can in principle be assembled from any set of FEM modes — but the $O(N^3)$ – $O(N^4)$ tensor is what rules it out at $N = 256$ in a plugin, and it is the reason the two cheap surrogates were chosen instead.

Finally, the mechanism of §5.1 is not new in synthesis: an amplitude-dependent tension computed from a global stretching estimate, fed back into the linear model’s tuning, is the

plate analogue of the tension-modulation models used for plucked strings, where the same first-order retuning reproduces the pitch drop of a hard pluck [37]. What is specific here is that the tension enters through the FEM sensitivities g_k of §3.2, so it costs one scalar per audio block on any drawn shape.

6 Implementation summary

Stage	Thread	Notes
Shape / boundary editing	message	ShapeCanvas ; outline + arc-parameterised segments
Mesh generation	message	milliseconds; preview in FemViewComponent
Eigensolution (6)	background	BackgroundTaskRunner ; progress; seconds
Model publication	message → audio	atomic pointer, generation-acknowledged reclamation
Filter bank (10)	audio	allocation-free; retunes throttled
Nonlinearities (16), (17)	audio	decimated γ update; bounded feedback

The numerical core is pure C++17 with no JUCE dependency and is validated by a console test suite; it lives in the **FxmeTools** submodule, split between **FxmeTools/acoustics** (**FemMesh**, **PlateModes**: the plate physics) and **FxmeTools/math** (matrix storage, Cholesky, subspace eigensolver: the part that has never heard of a plate), so that either half is reusable on its own.

References

- [1] A. W. Leissa, *Vibration of Plates*, NASA SP-160, Washington DC, 1969.
- [2] S. Timoshenko, S. Woinowsky-Krieger, *Theory of Plates and Shells*, 2nd ed., McGraw-Hill, 1959.
- [3] K.-J. Bathe, *Finite Element Procedures*, 2nd ed., Prentice Hall / K.-J. Bathe, 2014.
- [4] E. Catmull, R. Rom, “A class of local interpolating splines”, in *Computer Aided Geometric Design*, Academic Press, 1974, pp. 317–326.
- [5] A. Bowyer, “Computing Dirichlet tessellations”, *The Computer Journal* 24(2), 1981, 162–166.
- [6] D. F. Watson, “Computing the n -dimensional Delaunay tessellation with application to Voronoï polytopes”, *The Computer Journal* 24(2), 1981, 167–172.
- [7] L. S. D. Morley, “The triangular equilibrium element in the solution of plate bending problems”, *Aeronautical Quarterly* 19, 1968, 149–169.
- [8] R. Rannacher, “Nonconforming finite element methods for eigenvalue problems in linear plate theory”, *Numerische Mathematik* 33, 1979, 23–42.
- [9] I. Babuška, J. Osborn, “Eigenvalue problems”, in *Handbook of Numerical Analysis*, Vol. II, North-Holland, 1991, 641–787.
- [10] E. Cuthill, J. McKee, “Reducing the bandwidth of sparse symmetric matrices”, in *Proc. 24th National Conference of the ACM*, 1969, pp. 157–172.

- [11] D. A. Dunavant, “High degree efficient symmetrical Gaussian quadrature rules for the triangle”, *International Journal for Numerical Methods in Engineering* 21, 1985, 1129–1148.
- [12] K.-J. Bathe, E. L. Wilson, “Solution methods for eigenvalue problems in structural mechanics”, *International Journal for Numerical Methods in Engineering* 6, 1973, 213–226.
- [13] G. H. Golub, C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, 2013.
- [14] R. Courant, D. Hilbert, *Methods of Mathematical Physics*, Vol. I, Interscience, 1953.
- [15] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer, 1976.
- [16] R. L. Fox, M. P. Kapoor, “Rates of change of eigenvalues and eigenvectors”, *AIAA Journal* 6(12), 1968, 2426–2429.
- [17] J.-M. Adrien, “The missing link: modal synthesis”, in G. De Poli, A. Piccialli, C. Roads (eds.), *Representations of Musical Signals*, MIT Press, 1991, 269–298.
- [18] J. D. Morrison, J.-M. Adrien, “MOSAIC: a framework for modal synthesis”, *Computer Music Journal* 17(1), 1993, 45–56.
- [19] K. van den Doel, D. K. Pai, “Modal synthesis for vibrating objects”, in K. Greenebaum (ed.), *Audio Anecdotes III*, A. K. Peters, 2003.
- [20] S. Bilbao, *Numerical Sound Synthesis: Finite Difference Schemes and Simulation in Musical Acoustics*, Wiley, 2009.
- [21] R. Bristow-Johnson, “Cookbook formulae for audio EQ biquad filter coefficients” (the “Audio EQ Cookbook”), <https://www.w3.org/TR/audio-eq-cookbook/>.
- [22] A. Chaigne, C. Lambourg, “Time-domain simulation of damped impacted plates. I. Theory and experiments”, *Journal of the Acoustical Society of America* 109(4), 2001, 1422–1432.
- [23] N. H. Fletcher, T. D. Rossing, *The Physics of Musical Instruments*, 2nd ed., Springer, 1998.
- [24] T. von Kármán, “Festigkeitsprobleme im Maschinenbau”, *Encyklopädie der mathematischen Wissenschaften* IV/4, 1910, 311–385.
- [25] H.-N. Chu, G. Herrmann, “Influence of large amplitudes on free flexural vibrations of rectangular elastic plates”, *Journal of Applied Mechanics* 23, 1956, 532–540.
- [26] H. M. Berger, “A new approach to the analysis of large deflections of plates”, *Journal of Applied Mechanics* 22, 1955, 465–472.
- [27] A. H. Nayfeh, D. T. Mook, *Nonlinear Oscillations*, Wiley, 1979.
- [28] K. A. Legge, N. H. Fletcher, “Nonlinearity, chaos, and the sound of shallow gongs”, *Journal of the Acoustical Society of America* 86(6), 1989, 2439–2443.
- [29] M. Ducceschi, C. Touzé, “Modal approach for nonlinear vibrations of damped impacted plates: application to sound synthesis of gongs and cymbals”, *Journal of Sound and Vibration* 344, 2015, 313–331.
- [30] S. Bilbao, C. Webb, Z. Wang, M. Ducceschi, “Real-time gong synthesis”, in *Proceedings of the 26th International Conference on Digital Audio Effects (DAFx23)*, Copenhagen, 2023, pp. 1–8.

- [31] S. Bilbao, “Sound synthesis for nonlinear plates”, in *Proceedings of the 8th International Conference on Digital Audio Effects (DAFx-05)*, Madrid, 2005, pp. 243–248.
- [32] S. Bilbao, M. Ducceschi, F. Zama, “Explicit exactly energy-conserving methods for Hamiltonian systems”, *Journal of Computational Physics*, 2023.
- [33] Z. Wang, M. Puckette, “A fast algorithm for the inversion of the biharmonic in plate dynamics applications”, in *Proceedings of the 5th Stockholm Music Acoustics Conference*, Stockholm, 2023.
- [34] C. Touzé, O. Thomas, A. Chaigne, “Asymmetric non-linear forced vibrations of free-edge circular plates, part I: theory”, *Journal of Sound and Vibration* 258(4), 2002, 649–676.
- [35] O. Thomas, C. Touzé, A. Chaigne, “Non-linear vibrations of free-edge thin spherical shells: modal interaction rules and 1:1:2 internal resonance”, *International Journal of Solids and Structures* 42, 2005, 3339–3373.
- [36] Q. Nguyen, C. Touzé, “Nonlinear vibrations of thin plates with variable thickness: application to sound synthesis of cymbals”, *Journal of the Acoustical Society of America* 145, 2019, 977–988.
- [37] T. Tolonen, V. Välimäki, M. Karjalainen, “Modeling of tension modulation nonlinearity in plucked strings”, *IEEE Transactions on Speech and Audio Processing* 8(3), 2000, 300–310.
- [38] S. Bilbao, “A family of conservative finite difference schemes for the dynamical von Kármán plate equations”, *Numerical Methods for Partial Differential Equations* 24(1), 2008, 193–216.
- [39] C. Touzé, S. Bilbao, O. Cadot, “Transition scenario to turbulence in thin vibrating plates”, *Journal of Sound and Vibration* 331(2), 2012, 412–433.
- [40] M. V. Berry, “Regular and irregular semiclassical wavefunctions”, *Journal of Physics A: Mathematical and General* 10(12), 1977, 2083–2091.
- [41] R. H. Lyon, R. G. DeJong, *Theory and Application of Statistical Energy Analysis*, 2nd ed., Butterworth-Heinemann, 1995.
- [42] G. Zames, “On the input-output stability of time-varying nonlinear feedback systems — Part I”, *IEEE Transactions on Automatic Control* 11(2), 1966, 228–238.
- [43] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002.